Workshop in Machine Learning and Data Analysis: Project Report

Competition	March Machine Learning Mania 2025	
Track	Non-competitive	
Team	Gal Abuhatzira (abuhatzira@mail.tau.ac.il)	
Links	**GitHub Repository**	**Kaggle Notebook (includes demo)**
Final Kaggle Score	**0.09925** (1st place) with a "Kaggle winning strategy", and **0.10522** (3rd place) without it	

Introduction & Challenge:

In this challenge, we are given historical data about NCAA men's and women's regular season and tournament games, and we are required to predict the probability of Team A winning against Team B for every possible matchup between two teams A and B that could play each other in the 2025 tournament (both men and women). By convention, Team A is the team with the smaller TeamID between the two.

Kaggle then looks at the n matchups that actually took place during both tournaments, takes the relevant subset of our predictions to those matchups $p_1, \dots, p_n$ and the true outcomes $y_1, \dots, y_n$ (where $y_i = 1$ if Team A won, else $y_i = 0$), and evaluates the Brier score:

$$\text{score} = \frac{1}{n}\sum_{i=1}^{n}(p_i - y_i)^2$$

In our case, $n = 134$ (67 games in each tournament).

A few reasons that indicate the difficulty of the prediction are:

- **Data**: there are limited labeled tournament games (only 1383 for men and 895 for women) versus a huge number of possible matchups (131,408). I expand on this in the following sections.
- **March-Madness**: the prevalence of unpredictable "upsets", where a team with a low seed suddenly knocks out a top-seeded team, possibly due to key players getting injured/suspended/benched, having an off-night, matchup-specific effects, etc.

The Data:

The Kaggle dataset provides a wealth of historical information regarding regular season and tournament performance. This data can be organized into five sections:

1. **The Basics**: This section includes the final scores for all regular season and tournament games since the 1985 season (compact results). It also provides tournament seeds from that season onwards, which serve as a primary indicator of each team's relative strength entering the postseason.
2. **Team Box Scores**: This provides game-by-game statistics at the team level for both the regular season and the tournament (detailed results). Data is available starting from the 2003 season for men and the 2010 season for women. These metrics offer a significantly more informative layer of performance analysis compared to the simple final scores found in the basics section.
3. **Geography**: This section details the city and state locations for all regular season and tournament games since the 2010 season.
4. **Public Rankings**: For men's teams, this provides weekly rankings from dozens of top-tier rating systems (the Massey Ordinals) since the 2003 season.
5. **Supplements**: This includes additional supporting information, such as coaches for each team throughout the seasons.

The total data size is 183 MB across 36 csv files.

Data preprocessing:

The preprocessing of the data consisted of several stages, driven by a few key considerations:

- **Using only detailed results:** As discussed in the Data section, Kaggle provides two types of game-by-game results for both the regular season and the tournament: compact and detailed. The compact data contains a minimal set of columns: the winning and losing teams ID, the final scores, the winner's location (home/away/neutral), the number of overtimes, and the season and day number of the game. The main advantage of the compact data is its coverage across many more seasons: 1985-2025 for men and 1998-2025 for women. The downside is that this compact game-level information is often not rich enough to construct a large set of informative features, especially for the models I ended up using, such as XGBoost. In contrast, the detailed data includes every column found in the compact data plus many more box-score statistics, such as assists, blocks, steals, and rebounds. These capture different aspects of team performance, and I elaborate on how they translate into features in the Features section. The main caveat is that detailed data exists for fewer seasons: 2003-2025 for men and 2010-2025 for women. In earlier versions of the project, I experimented with combining compact and detailed data by training separate Logistic Regression models on each and blending their predictions. However, as I shifted to the final model set, the extra seasons provided by the compact data did not improve cross-validation results, while they significantly increased the complexity of the code. I discuss this further in the Models section. Overall, using the detailed data allowed me to build stronger, more indicative features and kept the training distribution closer to 2025, as NCAA basketball has changed substantially since 1985 due to factors such as rule changes, the "one-and-done" phenomenon, and the evolving level of competition. For these reasons, I chose to use only the detailed dataset in the final pipeline.

- **Overtime adjustment:** As mentioned, each game's statistics are reported for the entire game, which can be longer than 40 minutes if it goes to overtime. An NCAA game lasts **40 + 5·NumOT** minutes, where **NumOT** is the number of overtime periods. To normalize all statistics to a standard 40-minute game, for every relevant column **col** I applied:

$$col \leftarrow \frac{col}{\left(\frac{40 + 5 \cdot NumOT}{40}\right)}$$

  This prevents inflating a team's per-game statistics simply because they played more overtime minutes. For example, without this adjustment, a team might be mistakenly interpreted as scoring more than average when in reality they just happened to play more overtime games.
- **Transform each game into two rows**: The raw NCAA data stores each game in a winner/loser format, which is inconvenient for feature engineering since I want a consistent "team vs opponent" view. Therefore, I transformed every game into two rows: one from the winner's perspective, and one from the loser's perspective, while also fixing the location field so it still refers to team's location. This will help us make (Season, TeamID) aggregation correct later on, and compute season-level averages/sums with unified logic, while ensuring every game contributes information to both teams symmetrically.

Feature Engineering:

The strategy in this part was to extract as many meaningful features as possible from the data and then let permutation importance decide what actually helps the **Brier score** and what doesn't, while doing my best to avoid too much correlation between features.

Season-level features: The detailed data provides many useful box-score stats for each team in each **game**. The idea was to compute **season** statistics for each team, so we can summarize how well the team performed throughout the regular season before entering the tournament.

The simpler features are straightforward averages, like points scored/allowed per game (**Score**, **OppScore**), the difference between them (**RegDiff**), and win percentage (**W**). The more advanced features use statistics that are scaled by the number of possessions, since raw totals can be misleading if teams play at different paces. So first we estimate possessions:

Poss: the number of possessions a team had in a game. The commonly used formula is the following:

$$Poss = FGA + TO + 0.44 \cdot FTA - OR$$

- we also compute a few additional rate-based features. For example, assists per possession (**AST_Rate**) to capture how well a team moves the ball and generates good shot opportunities, fouls per possession (**PF_Rate**) as an indicator of defensive discipline or how often a team is forced into fouling, and steals per opponent possession (**STL_Rate**) to reflect how well a team creates turnovers on defense.

The main advanced features are the **"Four Factors"**, popularized by Dean Oliver in *Basketball on Paper* [1]. These are considered fundamental indicators of team success. We compute each stat per game and then take its mean over the regular season:

- **eFG (Effective Field Goal %)**: measures shooting efficiency while giving extra weight to 3PT shots. Teams with higher eFG tend to convert possessions into points more efficiently, which makes them more likely to outscore opponents and win.

$$eFG = \frac{FGM + 0.5 \cdot FGM3}{FGA}$$

- **TOV (Turnover rate / Turnover %)**: measures turnovers per possession. Since a possession can end with a made shot, a missed shot, or a turnover, high turnover rates mean fewer shot attempts and fewer scoring opportunities.

$$TOV = \frac{TO}{Poss}$$

- **ORB (Offensive Rebound %)**: measures how often a team grabs offensive rebounds out of all available rebounds on that end. High ORB% means extra possessions, which increases scoring chances and can decide close games.

$$ORB = \frac{OR}{OR + OppDR}$$

- **FTR (Free Throw Rate)**: measures how often a team gets to the free-throw line relative to field goal attempts. Getting to the line consistently usually indicates an offense that forces fouls and can generate "easy" points.

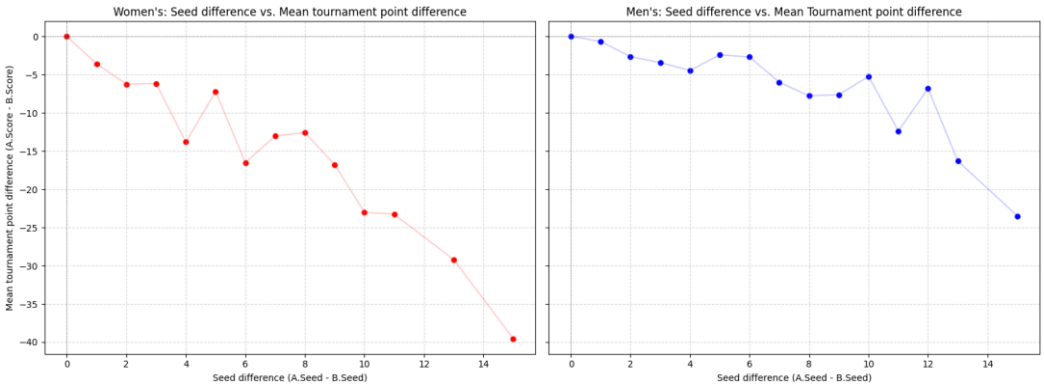
$$FTR = \frac{FTA}{FGA}$$

We also compute many features from a defensive perspective. Some worth mentioning are the opponent's eFG% from a defensive perspective (**Def_eFG**), i.e., how efficiently a team allows opponents to score. Lower values of this metric generally indicate stronger defense and better shot suppression. We also use block rate (**BLK_Rate**), which indicates what percentage of opponent shot attempts a team manages to block, effectively denying scoring opportunities.

In order to avoid correlation, I chose to keep just the advanced features listed above and not include the means features over plain columns in the detailed data CSV.

Seed feature: We also use the tournament **Seed** as a feature, directly from the seeds CSV Kaggle provides us. It's basically a direct indicator of how strong a team is relative to the other teams in the league. Lower seed numbers (like 1–4) generally correspond to stronger teams and historically win more often, so seed gives a strong sense for matchup outcomes.

In the following graphs, we can see that as the value of X grows larger (i.e., Team B finished the regular season with a better seed than Team A), Team B tends to win by a larger margin on average. The most important thing is that when we get to large seed differentials (around 8 and above), the women's win margins become much more decisive compared to the men's. This points to the different tournament structure in the women's bracket, with fewer truly elite teams and clearer favorites.



That was the main reason we introduced the **Seed_Gender_Interact** feature. The favorite in the women's tournament typically wins more often, so the same seed gap usually implies a larger win-probability shift in women's games than in men's games. This interaction lets the model learn a stronger seed effect specifically for women, instead of forcing one shared seed-to-win mapping for both genders.

Situational features: In the detailed CSV, Kaggle provides the specific day of the regular season each game was held (1 through 134) as well as the game's location. The idea was to leverage the fact that tournament games take place after the regular season

and capture a team's "momentum" heading into the postseason by calculating the **Late_WinPct** feature (win percentage in the last two weeks of the season). In addition, because the tournament is held on neutral courts, I wanted to capture how well a team performs away from home, leading us to calculate the **Road_WinPct** feature (win percentage in neutral or away games). However, Late_WinPct failed to appear in any of the three permutation importances analyses we conducted, likely due to a small sample size (often only 3–4 games) which makes it a noisy predictor compared to stable, full-season metrics. Road_WinPct underperformed in permutation importance as well, likely because of significant overlap with the EloEnd feature, which already captures the difference between home and away wins, a point I will expand on right away.

EloEnd feature: The **EloEnd** rating is the project's most important feature. Unlike a basic win percentage, this Elo system is iterative and self-correcting. The value of a win is determined by the opponent's quality at the time the game was played, the location, the point differential, and more. My implementation incorporates several advanced adjustments to ensure the final "End of Regular Season" rating is as predictive as possible of a team's true strength.

We iterate through the detailed data in chronological order, sorting by Season and DayNum. When encountering a match between Team A and Team B, we compute the expected win probability of Team A (E_A) using the standard Elo formula:

$$E_A = \frac{1}{1 + 10^{\frac{R_B - R_A}{400}}}$$

Before calculating this probability, we apply the following adjustments:

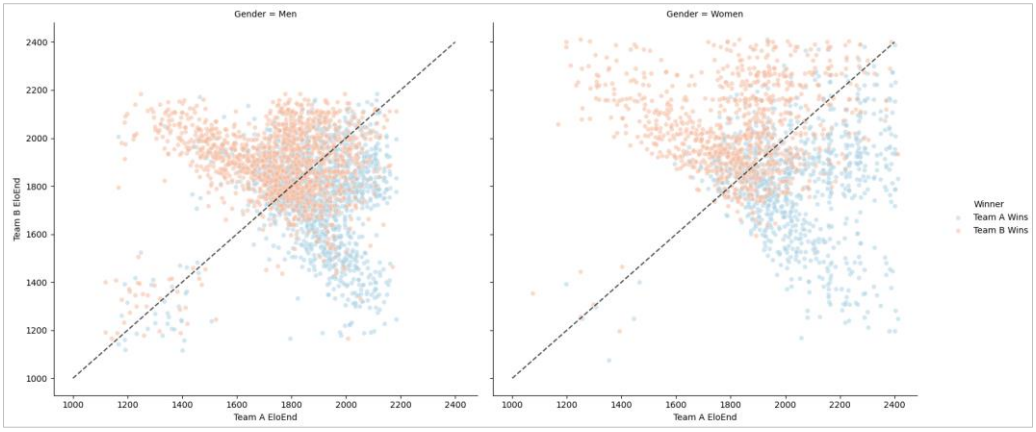
- **Home-Court Advantage (HCA)**: we apply a 75-rating-point adjustment in favor of the home team. This reflects the fact that playing at home is usually considered an advantage in sports. As a result, a home win is treated as less surprising, while an away win is treated as more impressive. This prevents the Elo system from over-rewarding teams for expected home wins or over-penalizing teams for road losses.
- **Margin of Victory (MOV) multiplier**: To distinguish between a close game and a blowout, we use a logarithmic multiplier based on the points differential. This prevents teams from artificially inflating their Elo score by dominating a weak opponent during "garbage time", while still rewarding truly dominant performances. We also implement a margin bound of 20, meaning a win by 30 or 40 points contributes the same to the rating as a 20-point win, as massive blowouts do not necessarily indicate a stronger team.
- **Early Season K-factor Scaling**: We implemented an early-season K-boost that increases the rating updates intensity by 50% at the start of each year. This multiplier linearly decays over the first 20 games, allowing the model to rapidly recognize current-year team strength before stabilizing into more conservative updates as the season progresses.
- **Mean Reversion**: For the first NCAA season in our data, teams started with a base Elo of 1500. Teams joining in subsequent seasons started at 1300. Between seasons, we retain an indication of the previous year's strength while accounting for the roster changes common in college sports (due to graduation, the NBA draft, and the "One-and-Done" phenomenon). A team's starting Elo for a new season is calculated as 70% of its final rating from the previous season plus 30% of the mean Elo rating of its conference. This allows us to track historical performance without being overly dependent on outdated rosters.

After these adjustments are applied, we calculate a delta value indicating the quality of the result. This delta is added to the winner's rating and subtracted from the loser's.

My Elo methodology is largely inspired by the *Silver Bulletin (SBCB)* system and *FiveThirtyEight* system [2], [3]. One feature I aimed to implement but ultimately excluded was **travel distance**. The *SBCB* system suggests that a team traveling more miles to a neutral site, or traveling at all for a road game, is less likely to win due to time-zone shifts and fatigue. While Kaggle provides geographic data, it does not include pre-calculated travel distances between cities. Although designing an estimation system is possible, I determined that the current Elo system already provides such high predictive power that the marginal gain from travel data was not necessary.

It is worth mentioning that established systems like the *Silver Bulletin (SB)* do not always disclose all of their exact parameters due to the competitive nature of sports betting. However, they did provide some, and I performed a grid search across the range of possible values they provided for each adjustment. The parameters stated above represent the optimal combination discovered through my testing and are the ones utilized in the final project.

In the following graphs, we can see that as the difference between the Elo ratings grows, the team with the higher rating becomes increasingly more likely to win the match. Additionally, we observe that "upsets", where a lower-rated team defeats a stronger opponent, are significantly more common in the men's tournament than in the women's. This further indicates our observation that there are fewer elite women's teams.



Quality feature: After trying the features listed above, experimenting with few other features, changing the simple Logistic Regression to an ensemble model, I wasn't able to improve the Brier score. Therefore, I took inspiration from *Raddar*'s notebook and implemented a "team quality" rating using a simple GLM on regular-season results [4]. The idea is to treat each regular season game as information about the relative strength of two teams and fit a model that explains point differential using only information about who those teams are (TeamID). Concretely, for each season (and separately for men/women), we fit a GLM of the form:

$$PointDiff \sim TeamID + OppTeamID - 1$$

This assigns each team a coefficient that acts like a latent strength parament: if Team A's coefficient is higher than Team B's, the model expects A to beat B by more points on average. After fitting, we center the quality values within each (Season, Gender) so the numbers are stable and comparable within that season. To keep the model focused and efficient, we focus on a relevant set of teams: tournament teams plus non-tournament teams that managed to beat a tournament team at least once, while all other teams collapsed into a single baseline category. This keeps GLM smaller, faster, and more focused on the tournament-level system, while still accounting for teams that interacted with it in meaningful games. Overall, this feature is basically a "strength rating" learned from regular season score margins, and it ended up being consistently high in permutation importances, so it gives a real signal beyond seeds and other features, almost the same importance as our EloEnd.

There were many features that weren't practical for various reasons or simply didn't improve the Brier score (and sometimes even made it worse). A few notable examples include:

- **Experience features:** One idea was to compute "tournament experience" for each team in a given season based on the previous few years: the seeds earned, the average seed over that span, the number of tournament wins achieved, and how often the team made a "deep run." The motivation was that a more experienced team might handle tournament-only scenarios (high pressure, short turnarounds, unfamiliar opponents) better and therefore perform more reliably. In practice, this feature family had two main issues. The primary problem was that their signal was mostly redundant with the Seed, EloEnd, and Quality features we already had. Strong teams tend to get better seeds and therefore naturally gain more tournament experience by playing against weaker teams and advancing further. Consequently, the experience variables didn't add much new information and, in some cases, added noise. There were also many policy questions, such as how to evaluate the average seed of a team that made the tournament only once or twice in the last five years. Furthermore, it was difficult to distinguish between teams that missed a 16-seed by a single game versus teams that weren't even close to qualifying. As a result, these features didn't improve cross-validation Brier scores and were removed from the final pipeline.

- **Massey Rankings**: One feature I was particularly looking forward to incorporating was *Kenneth Massey*'s public ranking data, as noted in my midpoint review. Massey provides weekly lists of rankings from many different systems, offering a comprehensive look at where various methodologists rank each team. The primary utility of this data is to observe where tournament-bound teams land in these power rankings during the final week of the regular season. However, a significant limitation is that these rankings are only available for men's teams and lack coverage for the women's tournament. I explored a naïve approach of incorporating the feature for men while filling the women's values with a default baseline, but this introduced noise and failed to improve the Brier score. I also considered splitting the dataset to train a men-only model with Massey features and a separate women's model without them. Ultimately, neither subset was large enough on its own to maintain the predictive power required for a competition of this scale. After testing several configurations, I concluded that since we only use the Massey data as a collection of end-of-season power rankings, our EloEnd and Quality features serve as excellent substitutes.

Validation methodology:

To ensure our models generalize to future tournaments and avoid overfitting, I implemented a Leave-One-Subject-Out (LOSO) cross-validation strategy. Given the nature of the NCAA tournament, where team dynamics, rosters, injuries, and rules vary significantly from one season to the next, I identified LOSO as the best validation strategy for this project. By iteratively holding out a single season as a validation set and training the models on every other available season in the dataset, we simulated the challenge of predicting an entire tournament without any prior knowledge of its participants or outcomes.

The main key of this strategy was the generation of Out-of-Fold (OOF) predictions. For every game in the historical training set, the prediction was produced by a version of the model that had never seen any data from that specific tournament season. This approach ensures that our predictions accurately reflect the model's true predictive power and its performance on entirely unseen data. These OOF results were used in the feature selection process, where I utilized permutation importance to prune features that failed to surpass a certain non-negative delta threshold in our validation metrics. This filtering of the features ensured that the final pipeline consisted only of high-signal features that remained robust across multiple tests.

Finally, these OOF predictions provided the essential **training data for the Stacking architecture**. To optimize the final ensemble, I used the OOF outputs from the three base models, which are discussed in detail in the next section, to train the Meta-Logistic Regression. By training the meta-model on the base models' mistakes, the ensemble learned which models to trust in specific competitive scenarios. This approach allowed the system to balance the stable linear trends of the Logistic Regression model with the non-linear complexity of the XGBoost models. This multi-layered validation framework was the key to breaking through the project's initial performance ceiling and achieving a highly competitive final Brier Score.

Models:

My modeling strategy evolved through several iterations as I attempted to break through the performance ceiling of the standard Logistic Regression (LR) approach. I eventually transitioned from a single-model baseline to a Stacked Ensemble that combines the strengths of linear stability (Logistic Regression) and non-linear complexity (XGBoost).

I began with two Logistic Regression models trained on the difference between Team A and Team B features (A - B). One model was trained on the compact Data (Men since 1985, Women since 1998), which offered a larger sample size but fewer statistics per game. The other was trained on the detailed Data (Men since 2003, Women since 2010), providing a richer feature set at the cost of historical depth. After training both, I performed cross-validation to find the optimal weighted average of their predictions. The best result from this basic ensemble was a Brier Score of **0.16950**, achieved with a weight distribution of 0.65 for the compact model and 0.35 for the detailed model. When applied to the 2025 Kaggle competition data, this basic model achieved a surprisingly strong score of **0.11174** (156th place).

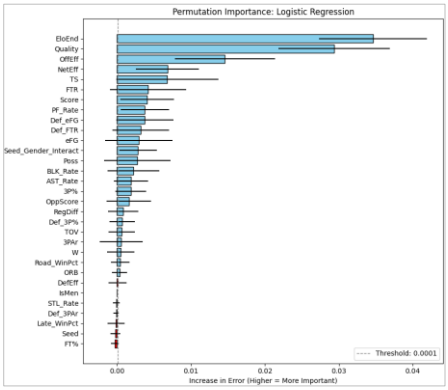
Despite the early success of the LR models, I eventually hit a ceiling where additional features and parameter tuning failed to improve the score. I concluded that a linear approach struggled to capture complex, non-linear interactions. To move past this, I turned to XGBoost:

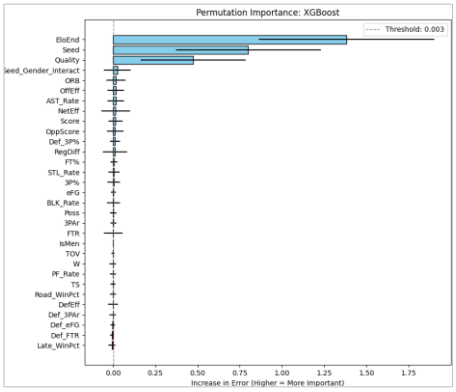
- **XGBoost Regressor (Margin-based)**: We train a regressor to predict the actual tournament point differential (**TourDiff**) between two teams. We then apply Platt Calibration to map these predicted margins into the win probabilities required for the competition.
- **XGBoost Classifier (Binary)**: We also train a direct binary classifier to predict win probability. This allows the model to focus purely on the win/loss outcome, preventing it from being skewed by massive "blowout" scores that might not accurately represent a team's true strength or dominance.

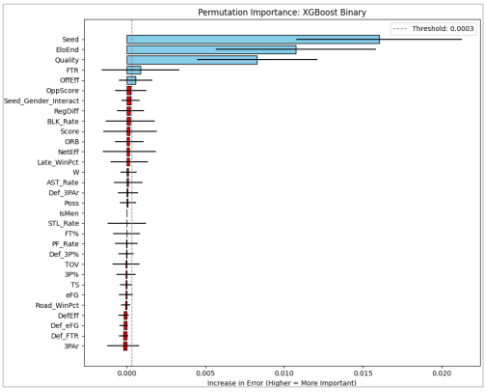
Because the original LR detailed model provided such high-quality results, I felt it was essential to include it in the final pipeline. The most critical leap in performance came from Stacking, based on the realization that the LR model and the two XGBoost models were wrong in different ways but often complemented each other.

- **OOF Stacking**: We use the OOF predictions from our three base models to train a final Meta-Logistic Regression. This meta-model learns which model to trust in specific scenarios based on historical performance.
- **Gender-Specific Meta-Heads**: During validation, I observed that feature values, specifically the impact of top-tier seeds (as said before), differ significantly between the Men's and Women's tournaments. To account for this, we fit two separate meta-models: one for the Men's tournament and one for the Women's. This allows the ensemble to automatically adjust its aggressiveness based on the gender of the matchup, leading to our most stable and competitive Brier Score.

Here we can see the results of the permutation importance for all 3 models:







The permutation importance analysis reveals a consistent hierarchy of predictive power across our three base models, with EloEnd, Quality, and Seed emerging as the dominant features. While the Logistic Regression model effectively utilizes a set of slightly less effective features, such as OffEff, NetEff, and TS, to refine its linear predictions, both XGBoost models are more concentrated on the top three metrics, and other variables provide only marginal improvements once core team-strength is accounted for. It is also important to note that the thresholds depicted in the importance graphs were determined empirically. I selected the specific cutoff for each model by identifying which threshold most effectively reduced the cross-validation Brier Score for each model.

Results:

Following our feature selection via permutation importance, we evaluate three distinct base models. While their individual scores were competitive enough relative to the models I started with, each model captured different nuances of the tournament's aspect.

- **Logistic Regression**: Interestingly, the LR model trained on detailed feature differences (A - B) emerged as the strongest individual baseline, achieving a Brier score of **0.16743**. Despite its relative simplicity, I recognized early in the project that this model handled the data effectively, yielding excellent results even as a standalone predictor. It was only after reaching an apparent performance ceiling with this architecture that I shifted the focus to XGBoost and its variants.
- **XGBoost Variants**: Both XGBoost configurations performed similarly, with the binary classifier achieving a slightly better Brier (**0.16861**) than the margin-based regressor (**0.16868**). The Mean-Absolute-Error (MAE) of **9.15129** points for the regressor indicates a high level of accuracy in predicting the actual spread of games before probability calibration.

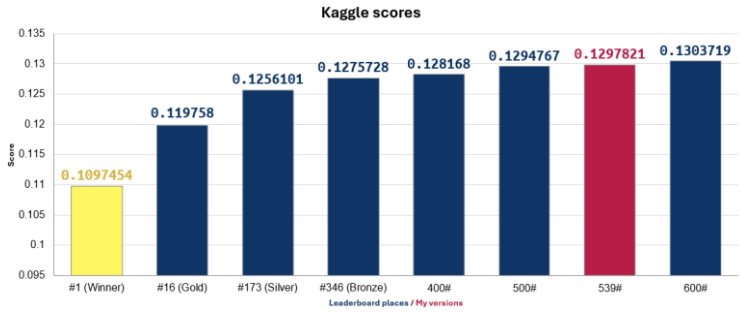
The final stage of the pipeline involves a Meta-Logistic Regression trained on the OOF predictions of the three base models. This ensemble was specifically designed to balance the stable trends of the Logistic Regression model with the complex, non-linear interactions captured by the XGBoost models.

While the Meta-Model's Brier score of **0.16764** is close to the standalone Logistic Regression baseline, the ensemble provides a critical layer of robustness. By weighing the three different perspectives from each individual model on team strength, the meta-model reduces the risk of failure in any single season.

This multi-layered approach of moving from raw data to engineered features, through permutation-based pruning, and finally into a gender-specific stacked meta-model, allowed me to produce a final submission that is both statistically sound and highly competitive within the tournament's high-variance environment.

Regarding the final competition results, the submission achieved a score of **0.10522**, which ranks this solution in the 3rd place for this completed tournament. Upon reviewing several published solutions from the top eight finishers, I observed that many competitors manually adjusted their model's probability output based on personal intuition or expertise - a Kaggle winning strategy. To test the potential impact of such heuristics, I examined a strategy of aggressive clipping. For example, by manually modifying the results, rounding any win probability above 80% up to 100% and any probability below 20% down to 0%, I aimed to minimize the Brier error on high-confidence games. When this strategy is applied, the score improves to a record best of **0.09925**. This hypothetical result actually surpasses the actual 1st place solution (**0.10411**), which also performed manual changes. Since this is a workshop project, I've decided to submit the solution that our model produced, without any manual modifications.

Note: Since the project submission deadline was delayed, I also had the opportunity to join the **2026** March Machine Learning Mania competition and submit a slightly less polished version of this project, switching the data to the 2026 competition. My solution finished in **539th** place out of 3,462 competitors, with a Brier score of **0.1297821**. The leaderboard results are shown below:



A likely reason for the gap between the 2025 and 2026 rankings is the high level of year-to-year unpredictability in this type of competition. A model that performs well on one tournament year is not guaranteed to perform equally well the following year, since the tournament outcomes depend on many noisy and difficult-to-predict factors. This can also be seen from the differences between the top-performing competitors across different competition years.

Lessons learned & future work:

In summary, this project revealed the infinite directions a machine learning project can take, from the initial selection of features and mathematical formulas appearing in basketball analytics literature, to the specific choice of parameters, models and validation frameworks. As stated in my original project proposal, I began this process from scratch, exploring the data through a simple Logistic Regression model. Although this model achieved surprisingly strong results in both validation and on the Kaggle leaderboard, I eventually reached a performance ceiling where additional features failed to move the needle.

The turning point occurred shortly after the midpoint review when I recognized the prevalence of XGBoost in top-tier notebooks solutions. Deciding to integrate XGBoost alongside my existing LR model unlocked new potential and allowed for much more sophisticated non-linear interactions. Thinking back, I believe the standalone Logistic Regression model might have achieved even greater heights had I implemented the refined feature engineering, such as the optimized EloEnd and Permutation Importance procedure, earlier in the development process. This suggests that even a simpler architecture can be remarkably effective.

However, many ideas did not yield the improvements I expected. For instance, several experimental features proved redundant because the EloEnd metric already captured the vast majority of the signal available in the Kaggle dataset. Furthermore, attempting to train separate base models for the Men's and Women's tournaments was unsuccessful. The resulting lack of training data for each individual subset led to weaker results.

I also explored my initial proposal's idea of annotating teams with external NBA career data, like the number of future star players or Hall of Famers a college produced. This was ultimately excluded from the final pipeline due to a lack of accessible historical datasets, coupled with the realization that individual professional success, often derailed by injuries or other external factors, might not serve as an indicative or reliable proxy for college-level team performance.

Future work could focus on engineering new features that capture signal currently missed by the Elo system, although this has proven to be a significant challenge. Additionally, exploring the inclusion of a third model type, such as a Neural Network, into our stacked ensemble could determine if a different perspective other than LR and XGBoost would further improve the final Brier Score.

References:

1. D. Oliver, Basketball on Paper: Rules and Tools for Performance Analysis. Dulles, VA: Potomac Books, 2004.
2. N. Silver, "How our SBCB ratings and NCAA tournament forecasts work", Silver Bulletin, Mar. 7, 2025. Available: https://www.natesilver.net/p/sbcb-methodology
3. N. Silver and R. Fischer-Baum, "How We Calculate NBA Elo Ratings", FiveThirtyEight, May 21, 2015. Available: https://fivethirtyeight.com/features/how-we-calculate-nba-elo-ratings/
4. Raddar, "vilnius ncaa", Kaggle Notebook. Available: https://www.kaggle.com/code/raddar/vilnius-ncaa